SIMATS ENGINEERING
Department of Computer Science and Engineering
ASSESSMENT TOOL 2 – CONCEPT MAPPING

Course Code:	CSA12	**Course Title:**	Computer Architecture
CO Assessed:	CO1 – Precision (BL4)	**Assessment:**	Concept Mapping
Weightage:	35%	**Total Marks:**	100
CO1 Statement:	Analyse the functional units of a computer system including CPU, memory, bus structures, and I/O components by examining instruction execution cycles, addressing modes, and performance metrics such as CPI, clock cycles, and benchmarking		
Student Name:	M.Koushik Reddy	**Reg. No.:**	192372185
Section / Batch:	CSE - AI	**Date:**	16-07-2026

Instructions to Students

- Identify the key computer architecture concepts, components, and performance parameters mentioned in the given topic or scenario.
- Organize the identified concepts logically by establishing relationships between CPU, memory, I/O devices, bus structures, instruction execution cycle, and performance metrics.
- Represent the concepts using an appropriate concept map with clear nodes, connecting lines, and meaningful linking phrases.
- Demonstrate the hierarchy and interdependence among concepts, showing how one concept influences or interacts with another.
- Ensure the concept map is complete, accurate, well-structured, and easy to understand by using appropriate terminology and logical connections.

QUESTIONS

1. A software company is upgrading its office computers and must decide whether to use ARM Cortex, Intel Core, or AMD Ryzen processors for different workloads. Construct a concept map linking these modern architectures with their instruction execution, cache usage, processing capability, and application areas. Show how their architectural features influence performance and energy efficiency. How do modern processor architectures differ in meeting the requirements of mobile, desktop, and high-performance computing systems?

2. A computer technician is troubleshooting a system that stores machine instructions in memory but displays memory addresses in a debugger. Construct a concept map linking Binary and Hexadecimal Number Systems with instruction representation, memory storage, CPU interpretation, and data conversion. Show how instructions are represented, stored, and interpreted during execution. How do binary and hexadecimal number systems simplify memory organization, instruction representation, and debugging?

3. A manufacturing company is upgrading its automated inspection system and notices delays when multiple sensors and storage devices communicate with the processor simultaneously. Construct a concept map linking CPU, Memory, I/O Devices, Single-bus, Multi-bus, and Hierarchical Bus structures, showing how data and control signals are transferred. Illustrate the effect of each bus organization on communication efficiency. How do different bus structures influence system performance and reduce communication bottlenecks?

4. A programmer is developing an industrial automation application and must choose the most suitable instructio[n] format and addressing mode for faster execution and efficient memory access. Construct a concept map linkin[g] Instruction Formats, Immediate, Direct, Register, Indirect, and Indexed Addressing Modes with Data Transfe[r,] Arithmetic, and Control Instructions. Show how each addressing mode affects execution speed and memory acces[s.] How do instruction formats and addressing modes influence program execution and processor efficiency?

5. An embedded systems engineer is comparing the 8085 and 8086 microprocessors for a factory monitoring syste[m] that requires efficient instruction execution and memory organization. Construct a concept map linking the instruction sets, register organization, addressing modes, and memory organization, including memo[ry] segmentation in 8086. Show how these features affect program execution and system performance. How do t[he] architectural differences between the 8085 and 8086 influence execution efficiency and memory management?

Code of Conduct

I KOUSHIK.M - 192372185 *(Name / Reg No) certify that this submission is my original work a[nd] that I have adhered to the guidelines specified for this assessment. I understand that any violation [of] academic integrity rules will result in disciplinary action.*

Signature of the Student: M.koushik Reddy

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

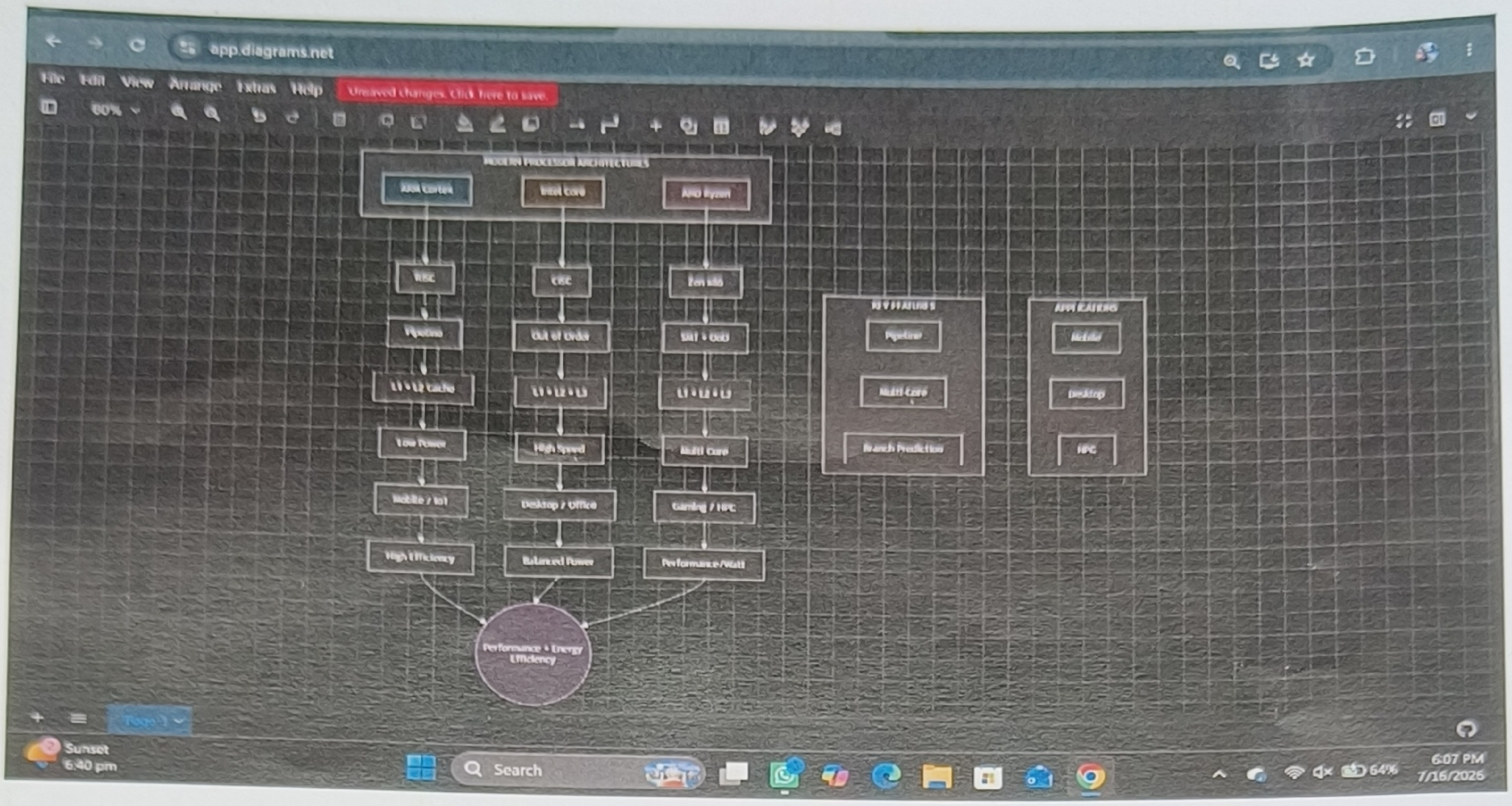
Marks Evaluation:

Q. No	Understanding of Problem Context (20)	Application of Concepts (30)	Problem-Solving Approach (20)	Accuracy of Solution (20)	Clarity (10)	Total (out of 100)
1	/ 4	/ 4	/ 4	/ 4	/ 4	
2	/ 4	/ 4	/ 4	/ 4	/ 4	
3	/ 4	/ 4	/ 4	/ 4	/ 4	
	/ 20	/ 30	/ 20	/ 20	/ 10	/ 100

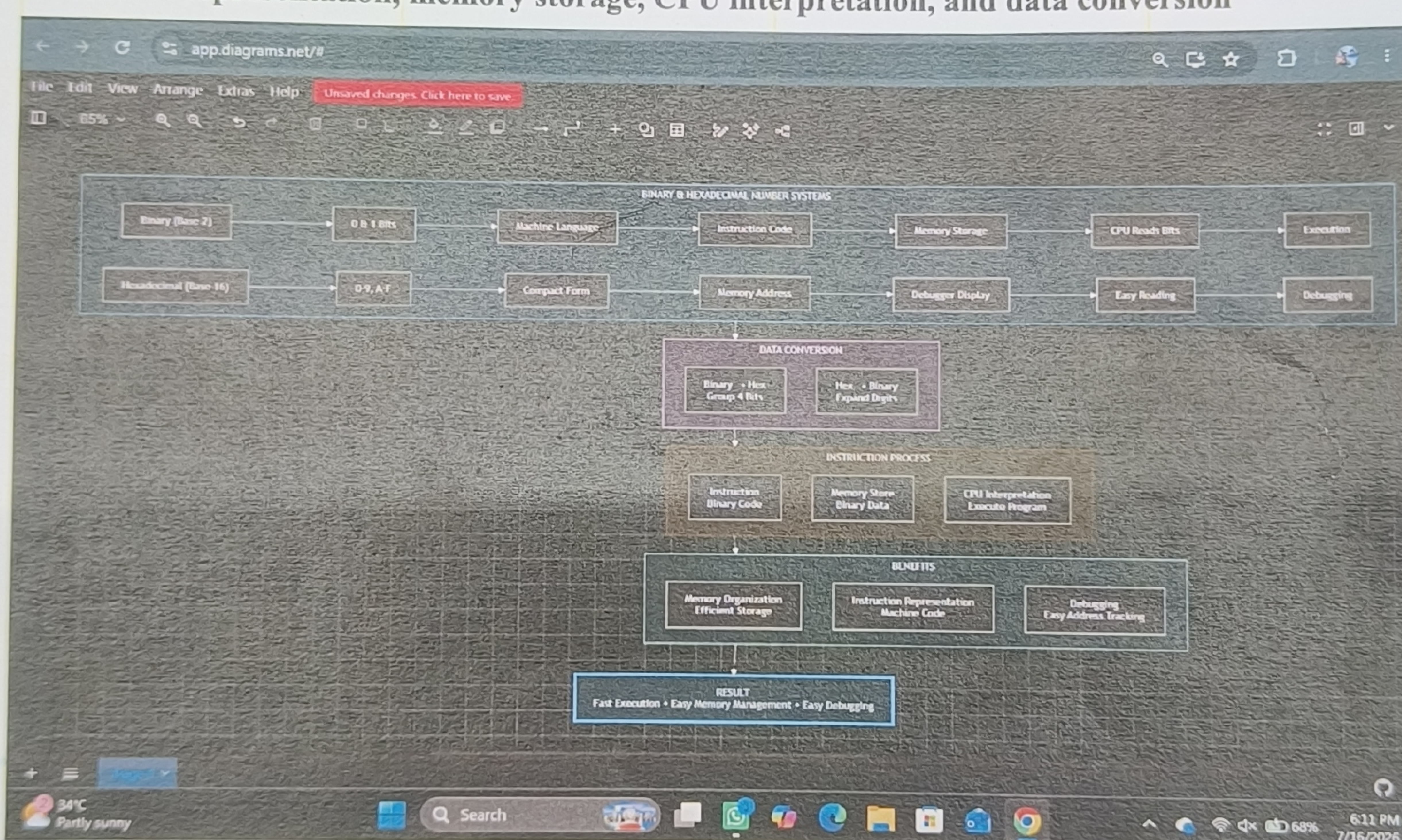
Course Coordinator

Faculty Incharge

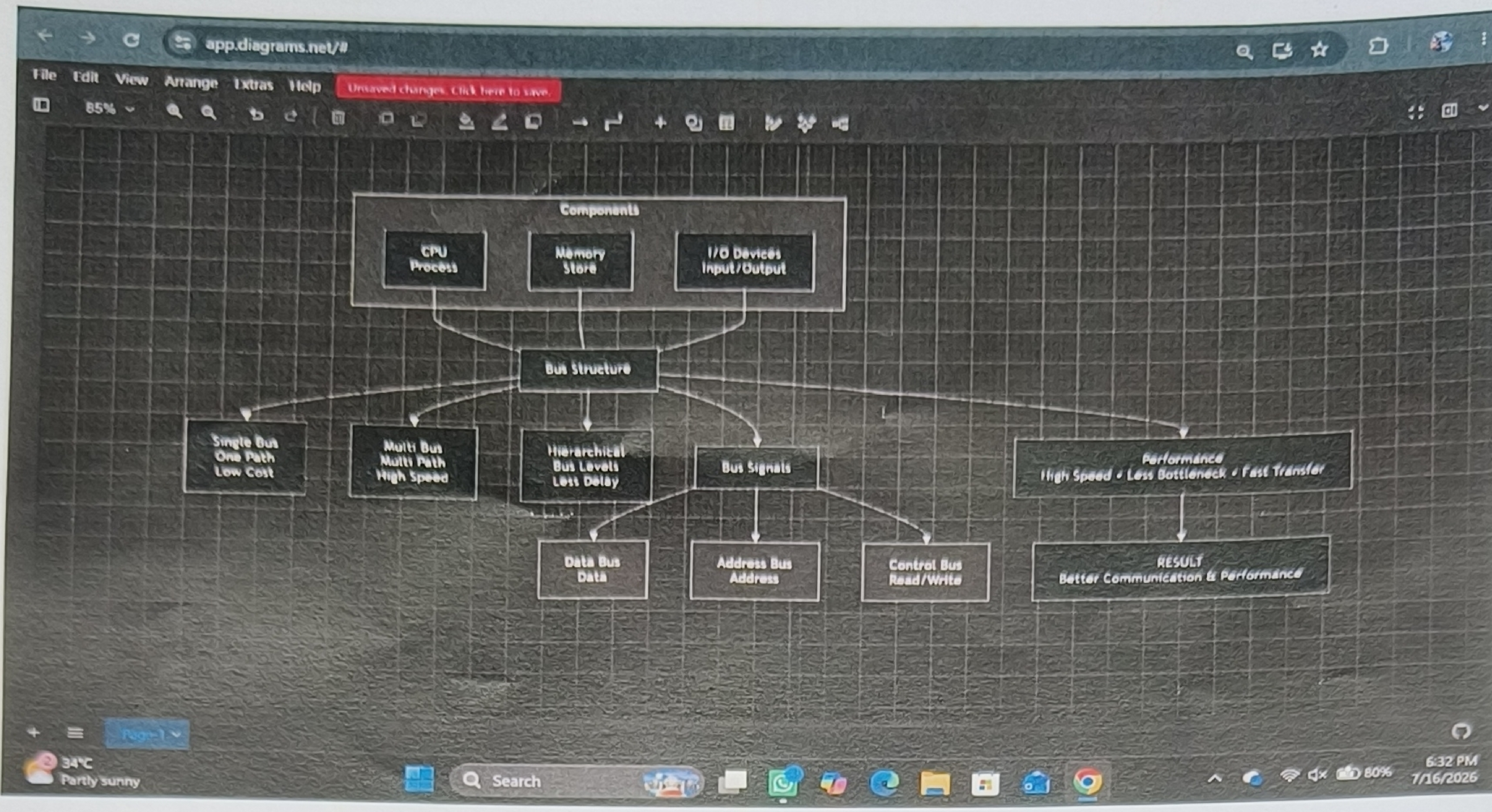
1.A software company is upgrading its office computers and must decide whether to use ARM Cortex, Intel Core, or AMD Ryzen processors for different workloads.



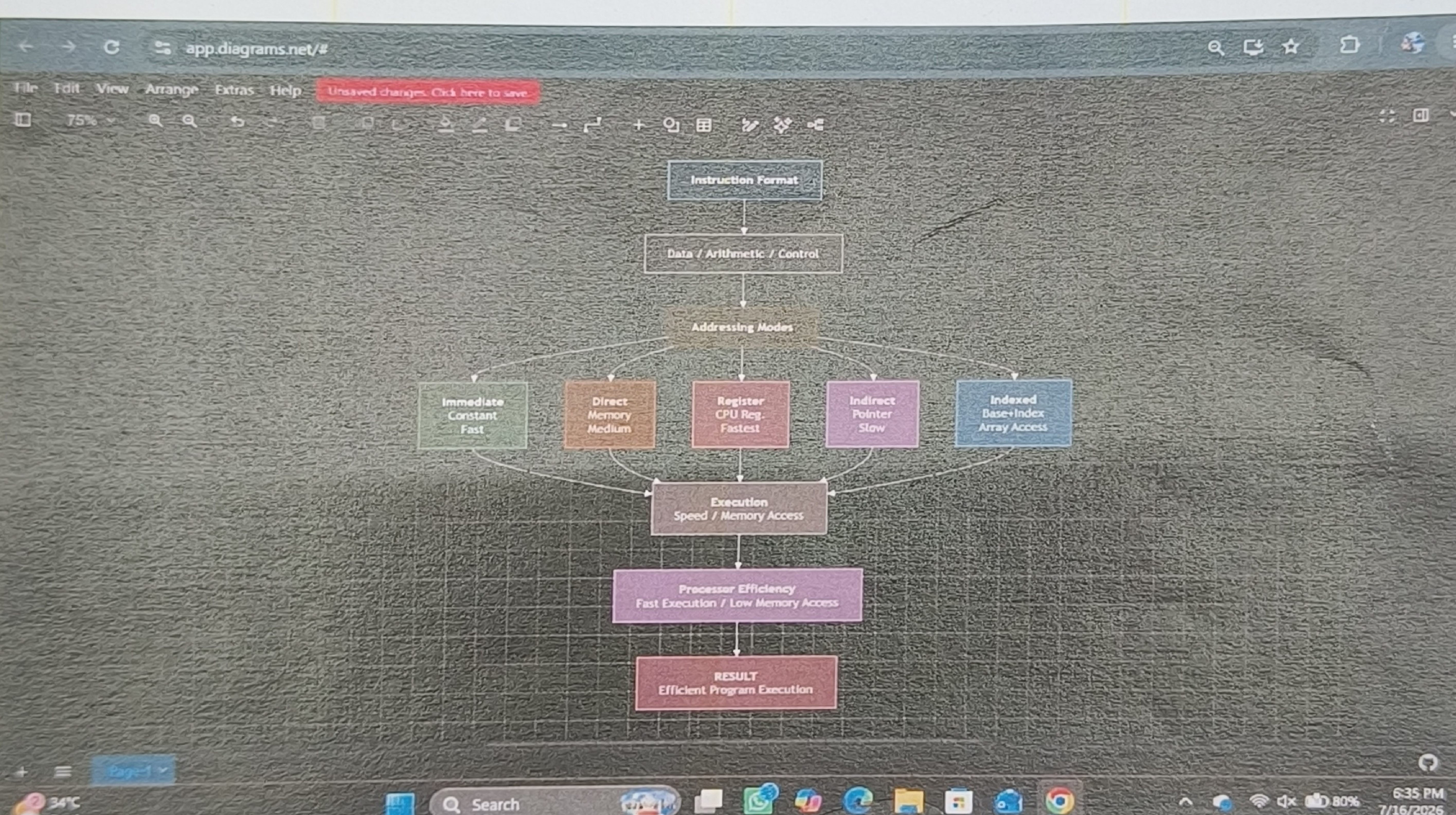
2. Construct a concept map linking Binary and Hexadecimal Number Systems with instruction representation, memory storage, CPU interpretation, and data conversion



3. Construct a concept map linking CPU,Memory, I/O Devices, Single-bus, Multi-bus, and Hierarchical Bus structures, showing how data and control signals are transferred.



4. Construct a concept map linking Instruction Formats, Immediate, Direct, Register, Indirect, and Indexed Addressing Modes with Data Transfer, Arithmetic, and Control.



5. How do the architectural differences between the 8085 and 8086 influence execution efficiency and memory management?